EXP 12: SECURITY AND GOVERNANCE IN ZOHO CREATOR

AIM

To study and apply the security and governance features of Zoho Creator, such as roles, permissions, data sharing, authentication, API security, auditing and change control, so that the Library Book Issue Management application and its data are protected and managed in a controlled way.

PROCEDURE

1. **Understand security and governance**
   - **Security** protects the application and its data from unauthorized access, misuse and loss.
   - **Governance** defines the rules, responsibilities and processes for how the application is built, changed, used and reviewed.
   - Both are applied to the library application, which stores member details, contact information and issue records.
2. **Control access using roles and permissions**
   - Open the application, go to **Settings** and then **User Permissions**.
   - Create permission sets (Librarian and Student) and set Access, View, Edit and Delete for each form, report and page.
   - Follow the **principle of least privilege**: give each user only the access needed for the work, for example read-only access for students.
3. **Define the role hierarchy and data sharing**
   - In the **Roles** tab, create the Librarian and Student roles, with Student reporting to Librarian.
   - Set **Share Data with Peers** to No, so that a student cannot see the records of other students.
   - Use the **Data Sharing** rules to give additional record access only where it is required.
4. **Protect sensitive data**
   - Make the important fields (Name, Email, Contact Number, Signature) mandatory and add field validations to avoid wrong data.
   - Restrict who can view or edit sensitive fields through the permission set, and hide fields that are not needed for a role.
   - Use the encryption option for sensitive fields, where it is supported by the plan.
5. **Secure user authentication**
   - Users log in through their Zoho account, so enable **two-factor authentication** and strong passwords.
   - Invite users only by email and remove the access of users who have left or no longer need the application.
   - Single sign-on and restricting access by IP address can be used where the plan supports them.
6. **Secure sharing and APIs**
   - Do not share forms or reports publicly unless it is required; prefer private links and login-based access.
   - In the permission set, check the **API and Security** permissions so that only the allowed roles can use the APIs.
   - Store credentials in **connections** and application variables instead of writing passwords or keys inside Deluge scripts.
7. **Enable monitoring and auditing**
   - Use the Deluge execution logs and the log details of workflows (as in the log workflow created earlier) to track what the scripts do.

- Use the audit history of records, where available, to know who created, edited or deleted a record and when.
- Review the logs regularly and investigate any failed or unusual activity.

8. **Apply governance rules**
   - Follow standard naming conventions for forms, reports and workflows (for example, On_IssueRequest_Submit_Workflow).
   - Use **approvals** and **blueprints** so that the issue process follows a fixed path: Requested, Approved, Book Issued and Book Returned.
   - Make changes first in Development, test them in Stage and then deploy to Production, and keep dated .ds backups of each version.
9. **Backup and recovery**
   - Export the data regularly in CSV or Excel format and export the application as a .ds file.
   - Archive old records as per the library retention policy and test that the backup can be restored.
10. **Review periodically**
    - Review the users, roles and permissions at regular intervals and remove unused workflows and old test data.
    - Check that the application follows the data protection rules of the institution and the data location policy of the Zoho data center.

Security Layers in Zoho Creator

Layer	Feature	Purpose
Authentication	Zoho login, two-factor authentication, single sign-on	Verify who the user is
Authorization	Permission sets, roles, data sharing	Control what the user can see and do
Data	Mandatory fields, validations, field permissions, encryption	Protect and keep the data correct
Integration	API permissions, connections, variables	Keep credentials and API access safe
Monitoring	Execution logs, audit history	Track activity and find problems

Governance Practices

Practice	Example in the library application
Least privilege	Students have read-only access to the book catalog and can only request issues
Change control	Development to Stage to Production deployment with .ds backups
Process control	Issue_Lifecycle_Blueprint and approvals for the issue process
Standards	Clear names such as Daily_Overdue_Reminder and Batch_Overdue_Fine_Calculation
Review	Regular check of users, roles, logs and unused workflows

RESULT

The security and governance features of Zoho Creator were studied and applied successfully. By using roles and permissions, data sharing rules, secure authentication, protected APIs and connections, logging, approvals and controlled deployment, the Library Book Issue Management application and its data were kept secure, and its development and use were managed in a controlled and well-documented way.